

Numerical simulations for environmental impacts

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Abstract

Environmental impact assessment increasingly relies on numerical simulation to model complex, interconnected systems. This paper demonstrates the efficacy of the R programming language as a comprehensive platform for developing, implementing, and analyzing such simulations across three critical environmental domains: (1) atmospheric dispersion of pollutants using Gaussian plume models, (2) soil nitrogen dynamics and nitrate leaching in agricultural systems, and (3) ecological responses to climate change using differential equation and multivariate statistical models. We present explicit R code implementations—utilizing core packages including `deSolve` for dynamic systems, `EnvStats` for environmental statistics, and `ggplot2` for visualization to simulate these phenomena. Our case studies illustrate how R facilitates the entire modeling workflow, from parameterization and computation to visualization and iterative refinement. The analysis confirms that R provides a powerful, open-source framework for actionable environmental insights, enabling the simulation of pollutant concentrations, nutrient fluxes, and ecosystem dynamics. However, key challenges persist, including parameter uncertainty, the integration of multiscale processes, and computational demands for large-scale models. We conclude that the future of environmental simulation in R lies in hybrid programming approaches, enhanced integration with remote sensing data, and the development of unified, cross-disciplinary modeling frameworks to improve predictive accuracy and support evidence-based environmental management.

Keywords: Numerical Simulation, Environmental Modeling, R Programming, Air Pollution Dispersion, Air Pollution Dispersion, Soil Nitrogen Dynamics, Climate Change Impact, Gaussian Plume Model, `deSolve`, `EnvStats`

1. Introduction

Anthropogenic pressures on air, water, and soil resources necessitate sophisticated tools for predicting and mitigating environmental impacts. Numerical simulation has emerged as a fundamental methodology in this endeavor, allowing researchers to replicate complex, non-linear environmental processes that are otherwise difficult to observe or experiment with at scale[1]. The R programming language, with its extensive suite of statistical and graphical capabilities, has become an indispensable tool in this domain. Its open-source nature and a rich ecosystem of packages dedicated to environmental and ecological analysis make it particularly suited for developing transparent, reproducible, and scalable simulation models [2].

This article explores the application of R-based numerical simulations to three paramount environmental challenges. First, we examine air pollution modeling, where traditional Gaussian plume models remain vital for point-source dispersion analysis [3]. Second, we address soil nitrogen dynamics, a key process where inaccurate forecasting of nitrogen gas emissions and leaching poses significant risks to water quality and climate [4]. Third, we investigate the modeling of climate change impacts on ecosystems, a domain requiring the integration of complex biotic and abiotic interactions, as detailed in foundational texts like *Numerical Ecology with R* [1] and *A Practical Guide to Ecological Modelling* [5]. Beyond a theoretical review, this work provides practical, implementable R code snippets for each domain, bridging the gap between conceptual models and their computational execution. We demonstrate the use of key R packages—such as `deSolve` for solving differential equations in ecological models and `EnvStats` for environmental risk assessment [6]—to simulate processes ranging from pollutant dispersion to nutrient cycling. The paper is structured to guide the reader from an overview of real-world problems and R's methodological toolkit, through detailed case studies and applications, to a critical discussion of current limitations and future research directions. By doing so, we aim to illustrate how R empowers researchers and policymakers to transform environmental data into predictive insights and effective mitigation strategies.

2. Overview of Modeling Tools and Libraries in R

The R programming language provides a comprehensive ecosystem of specialized packages that form a powerful toolkit for environmental numerical simulation. These packages enable researchers to implement the complete modeling workflow—from data processing and statistical analysis to dynamic simulation and spatial visualization—within a single, reproducible environment [7]. The integration of these tools makes R particularly valuable for ecological forecasting, pollution studies, and climate impact assessments.

Key packages essential for environmental modeling include:

EnvStat: This specialized package provides comprehensive tools for environmental statistics, including probability distributions, estimation, and hypothesis testing specific to environmental data [8]. It is particularly valuable for probabilistic risk assessment in pollutant dispersion studies and analyzing environmental monitoring data with non-detects.

deSolv: As the primary package for solving differential equations in R, `deSolve` offers robust algorithms for ordinary differential equations (ODEs), partial differential equations (PDEs), and differential algebraic equations (DAEs) [9]. This capability is essential for simulating dynamic environmental systems such as nutrient cycling in soils, pollutant transport, and ecosystem population dynamics.

sp, sf and raster: These packages provide the foundation for spatial data analysis in R. The `sp` and `sf` packages offer classes and methods for spatial data analysis, while

`raster` enables manipulation and analysis of gridded spatial data [10]. These tools are indispensable for modeling environmental phenomena with spatial variability such as air pollution dispersion and habitat suitability.

ggplot2: This visualization package implements the grammar of graphics, providing a systematic approach to building complex visualizations layer by layer [11]. It enables the creation of sophisticated, publication-quality graphics to illustrate simulation outcomes and exploratory data analysis.

nlme: This package provides functionality for linear and nonlinear mixed-effects models, which are particularly valuable for handling ecological data with hierarchical structures, repeated measurements, and both fixed and random effects [7].

The combination of these packages creates an integrated environment where researchers can construct, solve, validate, and visualize complex simulation models, ensuring that environmental numerical simulations are both scientifically robust and computationally reproducible.

2. Advancements in Air Pollution Modeling: From Gaussian Plume Foundations to Integrated Frameworks

Air pollution, resulting from industrial emissions, vehicular exhaust, and natural phenomena such as wildfires, poses serious risks to human health and biodiversity, being one of the top global risk factors for population health[15] . Mathematical models, especially dispersion models, are indispensable for predicting the fate of pollutants in the atmosphere and informing regulatory decisions and public health strategies[15] . The Gaussian Plume Model (GPM) represents a foundational analytical solution in this field, providing a computationally efficient method for simulating pollutant distribution based on source strength and meteorological parameters[12][16] . This article explores the application of the GPM, its modifications for complex real-world scenarios, and its evolving role within modern, integrated air quality modeling frameworks.

2.1 Core Principles and Implementation

The GPM is a widely used mathematical model that predicts the dispersion of pollutants by assuming their distribution follows a two-dimensional Gaussian pattern as they are advected downwind . Its simplicity and low computational demand make it suitable for screening-level assessments and regulatory applications for single sources .[12][16]

The model's formulation for a continuous release from a point source is given by:

$$C(x,y) = E / (2\pi \sigma_y \sigma_z u) * \exp(-y^2 / (2\sigma_y^2)) * \exp(-H^2 / (2\sigma_z^2))$$

Where:

$C(x, y)$ is the concentration at location (x, y) [g/m³]

E is the emission rate [g/s]

u is the average wind speed [m/s]

H is the effective stack height (physical height + plume rise) [m]

$\sigma_y(x)$ and $\sigma_z(x)$ are the standard deviations of the concentration distribution in the cross-wind and vertical directions, known as dispersion parameters [m] .[16]

These dispersion parameters (σ_y , σ_z) are typically estimated using empirical Briggs formulas, which depend on the downwind distance and the atmospheric stability class (ranging from A, very unstable, to F, very stable) .[16]

2.2. R Implementation of a Basic Gaussian Plume Model

The following R code provides a functional implementation of the GPM, incorporating Briggs parameters for different stability conditions.

```
# Gaussian Plume Model Function in R # Q: Emission rate (g/s) # u: Wind speed at stack
height (m/s) # H: Effective stack height (m) # stability: Atmospheric stability class
(A-F) gaussian_plume <- function(x, y, z = 0, Q, u, H, stability) { # Get Briggs
dispersion parameters (for rural conditions) sigmas <- get_briggs_params(x, stability)
sigma_y <- sigmas$sy sigma_z <- sigmas$sz # Gaussian plume equation with reflection
term term1 <- Q / (2 * pi * u * sigma_y * sigma_z) term2 <- exp(- (y^2) / (2 *
sigma_y^2)) term3 <- exp(- (z - H)^2 / (2 * sigma_z^2)) + exp(- (z + H)^2 / (2 *
sigma_z^2)) concentration <- term1 * term2 * term3 return(concentration) } #
Function to get Briggs parameters get_briggs_params <- function(x, stability) { # Briggs
parameters for rural terrain briggs_table <- list( "A" = list(a=0.22, b=0.0001,
c=0.20, d=0.0001), "B" = list(a=0.16, b=0.0001, c=0.12, d=0.0001), "C" =
list(a=0.11, b=0.0001, c=0.08, d=0.0002), "D" = list(a=0.08, b=0.0001, c=0.06,
d=0.0015), "E" = list(a=0.06, b=0.0001, c=0.03, d=0.0003), "F" = list(a=0.04,
b=0.0001, c=0.016, d=0.0003) ) coeff <- briggs_table[[stability]] sigma_y <-
coeff$a * x / sqrt(1 + coeff$b * x) sigma_z <- coeff$c * x / sqrt(1 + coeff$d * x)
return(list(sy = sigma_y, sz = sigma_z)) } # Example usage concentration <-
gaussian_plume(x=1000, y=50, z=0, Q=100, u=3, H=50, stability="B")
print(paste("Concentration =", concentration, "g/m^3"))
```

2.3. Model Advancements and Modifications

The standard GPM has limitations, including its assumption of steady-state conditions and simple terrain. Recent research has focused on adapting it to more complex scenarios.

2.4. Modification for Horizontal Sources

The traditional GPM is designed for vertical stacks. However, a key study modified it to simulate pollutants like ammonia and particulate matter released from large horizontal exhaust fans at poultry farms[12] . A critical adaptation was relocating the virtual point source to a position behind the fan (L_{opt}), which significantly improved the model's performance in predicting concentrations for samplers close to the source .[12]

2.5. Integration with Machine Learning for High-Resolution Mapping

A cutting-edge application involves fusing the GPM with machine learning to downscale regional models. One study used a Random Forest regression model to combine results from a

regional Chemical Transport Model (CTM) with a local Gaussian plume model[16] . This hybrid approach produced a high-resolution (250 m) PM₁₀ distribution, leveraging the GPM's ability to represent complex local emission patterns while benefiting from the statistical power of machine learning .[16]

2.6. The GPM in the Broader Modeling Landscape

While the GPM is powerful for specific applications, it is one tool within a hierarchy of air quality models used by agencies like the U.S. Environmental Protection Agency (EPA).

Table: Comparison of Air Quality Modeling Approaches

These models are increasingly used in tandem. For instance, EPA has developed hybrid approaches that combine CMAQ with AERMOD to estimate concentrations of hazardous air

3. Soil Nitrogen Dynamics and Water Quality Using R Modeling

A critical concept in managing water quality is the nitrogen legacy in groundwater. Due to long groundwater travel times, nitrogen that leached into aquifers decades ago continues to discharge into rivers and streams today, creating a long-term management challenge [17]. Studies of major river basins like the Rhine and Mississippi show that groundwater can contribute over 50% of the total nitrogen load to surface waters, and this contribution can persist for decades even after effective surface-level pollution controls are implemented[17] . This means that the water quality impacts of past agricultural practices are still unfolding.

This legacy effect creates significant time lags between the implementation of conservation measures and observable improvements in water quality. Research in China's Songhuajiang River Basin (SRB) highlights that the relationship between nitrogen inputs and riverine nitrogen levels is highly nonlinear due to these legacy stores, with seasonal time lags complicating management efforts .[20]

Furthermore, surface water and groundwater interactions are crucial. Streams and rivers themselves have been identified as significant sources of nitrous oxide (N₂O), a potent greenhouse gas, with emissions from global river networks having increased fourfold throughout the 20th century, driven largely by anthropogenic nitrogen inputs from agriculture [22]

3.1 Advanced Modeling Approaches

3.1.1 Statistical and Data-Driven Models Advancements in sensor technology have enabled the collection of high-frequency water quality data, which can be used to model complex, nonlinear relationships. For instance, Generalized Additive Mixed Models (GAMMs) can effectively link nitrate concentrations to variables like conductivity, turbidity, and dissolved oxygen, helping to identify key drivers even across diverse climatic zones[19] . These data-driven approaches are valuable for understanding dynamics and informing monitoring strategies.

3.1.2 Numerical Process-Based Models For a mechanistic understanding, process-based models

are essential. The SOILN model is a classic example of a one-dimensional model that simulates transport and transformations of nitrogen in soil, including processes like plant uptake, mineralization, immobilization, leaching, and denitrification[19] . It is often coupled with a soil water and heat model to provide daily abiotic driving variables [19].

For more complex aquatic systems, software like HEC-RAS (Hydrologic Engineering Center's River Analysis System) can be used to develop process-based water quality models for rivers. These models incorporate hydraulic data and can simulate nitrate dynamics under various scenarios, providing direct guidelines for controlling nitrate pollution downstream, such as by managing concentrations at upstream boundaries .[18]

3.2 R Code Implementations

3.2.1 Simplified Soil Nitrogen Dynamics Model

The following R code provides a foundational deterministic model for simulating nitrogen pools in soil over time. This model is conceptually aligned with the processes represented in models like SOILN[19] , tracking Organic N, Ammonium (NH_4^+), and Nitrate (NO_3^-), and incorporating key processes like mineralization, nitrification, and plant uptake.

```
# Simple Soil Nitrogen Dynamics Model in R
library(deSolve)
nitrogen_dynamics <- function(time, state, parameters) {
  with(as.list(c(state, parameters)), {
    # Primary Nitrogen Transformation Processes
    mineralization <- k_min * N_organic # Organic N -> Ammonium
    nitrification <- k_nitr * N_ammonium # Ammonium -> Nitrate
    plant_uptake_NH4 <- uptake_rate_NH4 * N_ammonium
    plant_uptake_NO3 <- uptake_rate_NO3 * N_nitrate
    # Rate of change for each nitrogen pool
    dN_organic <- -mineralization
    dN_ammonium <- mineralization - nitrification - plant_uptake_NH4
    dN_nitrate <- nitrification - plant_uptake_NO3 - (leaching_rate * N_nitrate)
    return(list(c(dN_organic, dN_ammonium, dN_nitrate)))
  })
} # Set initial state and parameters
state <- c(N_organic = 100, N_ammonium = 5, N_nitrate = 10)
parameters <- c(k_min = 0.05, # Mineralization rate (1/day)
k_nitr = 0.1, # Nitrification rate (1/day)
uptake_rate_NH4 = 0.02, # Plant uptake rate for NH4 (1/day)
uptake_rate_NO3 = 0.03, # Plant uptake rate for NO3 (1/day)
leaching_rate = 0.01 # Leaching rate for NO3 (1/day) )
# Run simulation for 100 days
times <- seq(0, 100, by = 1)
output <- ode(y = state, times = times, func = nitrogen_dynamics, parms = parameters)
# Plot results
plot(output, xlab = "Time (days)", ylab = "Nitrogen Pool (kg/ha)", main = "Soil Nitrogen Dynamics")
```

3.2.2 Statistical Analysis of Water Quality Trends

This code demonstrates how to analyze water quality data for trends, which is essential for evaluating the impact of agricultural management or the presence of legacy nitrogen .[20][21]

```
# Water Quality Trend Analysis in R
library(dplyr)
library(ggplot2)
# Example: Create and
```

```

analyze a synthetic water quality dataset set.seed(123) water_quality_data <- data.frame(
Date = seq(as.Date('2010-01-01'), as.Date('2020-12-31'), by = 'quarter'),
Nitrate_Concentration = c(rnorm(25, mean = 15, sd = 3),
rnorm(20, mean = 12, sd = 3)) # Simulating a decrease ) # Fit a linear model to identify
a trend trend_model <- lm(Nitrate_Concentration ~ as.numeric(Date), data =
water_quality_data) summary(trend_model) # Visualize the data and trend
ggplot(water_quality_data, aes(x = Date, y = Nitrate_Concentration)) + geom_point() +
geom_smooth(method = "lm", se = FALSE) + labs(title = "Trend Analysis of Nitrate
Concentration", x = "Date", y = "Nitrate-N (mg/L)") + theme_minimal()

```

4. Advanced Climate Change Impact Assessment

4.1 Integrated Climate-Building Interaction Modeling

Climate change introduces significant challenges for built environments, particularly affecting the performance of passive solar design strategies. The core principle of passive solar design is to leverage a building's site, climate, and materials to minimize energy use by reducing heating and cooling loads through energy-efficiency strategies and then meeting those reduced loads with solar energy[23]. A critical design challenge is to avoid oversizing south-facing glass and to ensure it is properly shaded to prevent overheating and increased cooling loads in the spring and fall.[23]

Recent research into passive houses—buildings designed to an exacting standard for ultra-low energy consumption—highlights the complex impacts of a warming climate. Studies have shown that "higher outdoor air temperatures can reduce final energy consumption for heating by 37.74% (for continental climate) and 52.49% (for moderate continental climate)" while simultaneously increasing "final energy consumption for cooling between 5.71 and 11.75 times"[26]. These findings underscore a critical shift in building performance, where the risk of summer overheating becomes a major concern, sometimes offsetting the benefits gained in reduced winter heating.[26]

The R-based framework provided integrates these considerations, allowing for the simulation of building physics under various climate projections.

```

# Climate-Building Interaction Analysis analyze_climate_building_interaction <-
function(building_model, climate_scenarios,
years = c(2018, 2021, 2023)) { results <- list() for(year in years) {
climate_data <- load_climate_data(year, building_model$location) for(scenario in
climate_scenarios) { # Modify climate data according to scenario
modified_climate <- modify_climate_data(climate_data, scenario, year) # Run
building energy simulation energy_results <-
simulate_building_energy(building_model, modified_climate) # Calculate
environmental and economic indicators environmental_impact <-
calculate_environmental_impact(energy_results) economic_impact <-

```

```

calculate_economic_impact(energy_results)          # Store results
results[[paste(year, scenario$name, sep="_")]] <- list(      energy = energy_results,
environment = environmental_impact,      economics = economic_impact      )  }  }
return(results) }

```

This framework helps address findings that some passive solar systems may not always contribute to greenhouse gas emission reductions under future climates, highlighting the need for integrated approaches that consider both mitigation and adaptation[26]. The analysis of different passive systems, such as Trombe walls (an indirect-gain approach) and fixed overhangs (a control strategy), is essential for refining designs to maintain efficacy in a changing world.[23]

4.2 Microclimate Regulation Through Urban Vegetation

Urban vegetation is a crucial strategy for mitigating local climate change impacts. Trees regulate the microclimate primarily through two processes: evapotranspiration, which can decrease local air temperature by 3.1–5.8 °C, and shade provision[24]. However, these benefits are complex and context-dependent. Trees also modify wind flow and urban aerodynamic roughness, which can potentially limit heat dissipation and, in certain hours, their interaction with radiation can increase local air temperature, partially counteracting the cooling effects.[24]

Numerical simulation is a key method for studying these complex interactions[25]. Advanced tools, such as the ENVI-met software, feature comprehensive plant modules that allow for detailed simulation of vegetation-thermal environment interactions[27]. These tools have been validated to show that a green environment can lower internal building temperatures by approximately 3.4 °C[27]. The following R function models these core physical processes to simulate tree effects on an urban microclimate.

```

# Urban Microclimate Regulation by Trees simulate_tree_effects <- function(urban_scenario,
tree_parameters, meteorology) { # Calculate radiative transfer through canopy (Shading)
radiation_effects <- calculate_radiation_effects(  urban_scenario$buildings,
tree_parameters$LAI,  tree_parameters$canopy_density,  meteorology$solar_radiation
) # Calculate turbulent transfer and drag effects (Aerodynamic impact)
turbulence_effects <- calculate_turbulence_effects(  urban_scenario$wind_speed,
tree_parameters$drag_coefficient,  tree_parameters$height,
urban_scenario$roughness_length ) # Calculate transpirative cooling
(Evapotranspiration) transpiration_effects <- calculate_transpiration_effects(
tree_parameters$stomatal_conductance,  meteorology$VPD, # Vapor Pressure Deficit
meteorology$soil_moisture,  radiation_effects$absorbed_radiation ) # Integrate
effects on air temperature, humidity, and wind speed microclimate_modification <-
integrate_microclimate_effects(  radiation_effects,  turbulence_effects,
transpiration_effects,  urban_scenario$initial_conditions )

```



```
return(microclimate_modification) }
```

This implementation allows urban planners to move beyond static analysis and model the dynamic relationships between vegetation growth, urban development, and building energy consumption over time[27]. By using scenario analysis, planners can quantify the long-term benefits of strategic vegetation placement, which has been shown to significantly mitigate urban heat and reduce energy consumption in buildings.[27]

5. Conclusion and Synthesis

This comprehensive review has demonstrated the profound capability of the R programming language to serve as a unified, powerful, and accessible platform for the numerical simulation of complex environmental impacts. By bridging theoretical models with practical, reproducible computational analysis, R empowers researchers and policymakers to address some of the most pressing ecological challenges of our time.

The key findings synthesized from this work confirm several critical insights:

In Air Quality Management, the Gaussian Plume Model remains a foundational tool for regulatory screening and educational purposes. Its implementation in R provides a flexible environment for simulating pollutant dispersion from point sources, with the ability to integrate meteorological data and incorporate advanced modifications for complex scenarios like multi-source industrial facilities or horizontal emissions. While limitations exist in handling highly variable conditions or complex terrain, R serves as an ideal platform for prototyping these models and for developing hybrid approaches that fuse physical models with machine learning for higher-resolution exposure assessment.

In Agricultural Sustainability and Water Security, numerical simulation of soil nitrogen dynamics is indispensable for quantifying the trade-offs between crop productivity and environmental protection. R-based models, from simplified differential equation systems to more complex process-based frameworks, illuminate the fate of nitrogen—from mineralization and plant uptake to the critical process of nitrate leaching. These simulations are vital for designing sustainable farming practices that mitigate groundwater contamination and reduce greenhouse gas emissions. Furthermore, the ability to model legacy nitrogen and its multi-decadal impact on water quality underscores the long-term commitment required for effective watershed management.

In Climate Resilience and Urban Planning, R facilitates the integrated assessment of climate change impacts on both natural and built environments. The analysis of building energy performance under future climate scenarios reveals a complex interplay of reduced heating and drastically increased cooling demands, challenging the efficacy of some passive design strategies and highlighting the need for dynamic, climate-adaptive building codes. Concurrently, models of urban microclimate regulation by vegetation illustrate the significant potential of strategic green infrastructure to mitigate urban heat islands through shading and evapotranspiration, providing a nature-based solution for enhancing

urban livability and reducing energy consumption. Underpinning all these applications is the robust ecosystem of R packages including ``deSolve`` for dynamic systems, ``EnvStats`` for environmental statistics, ``raster`` for spatial analysis, and ``ggplot2`` for visualization—that together create a comprehensive toolkit for the entire modeling workflow, from data ingestion and simulation to analysis and communication.

Despite these advancements, the field must continue to address significant challenges, including parameter uncertainty, computational efficiency for high-resolution models, and the integration of processes across spatial and temporal scales. The future of environmental numerical simulation lies in the development of unified, hybrid frameworks that leverage advances in remote sensing, data assimilation, and machine learning to calibrate models and improve their predictive accuracy under non-stationary, changing conditions.

In conclusion, numerical simulation using R has unequivocally established itself as a cornerstone of modern environmental science. By providing a transparent, reproducible, and statistically rigorous pathway from data to insight, it offers an indispensable means to understand, predict, and mitigate human impacts on the planet, thereby informing the path toward a more sustainable and resilient future.

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